Forecasting Influenza Cases in Australia Using SARIMA and Exponential Smoothing Models

Natalie Chidikwindi

1 Introduction

Influenza is a respiratory illness characterised by symptoms such as fever, headache, sore throat and cough. It spreads rapidly and causes seasonal epidemics, which can increase mortality and place additional pressure on healthcare systems. Identifying periods during which influenza cases are likely to increase can help healthcare providers and the public prepare more effectively.

Forecasting future influenza cases can support this planning by providing estimates of expected case numbers over time. Seasonality is particularly important in influenza dynamics because case numbers tend to rise during particular periods of the year. Accurate forecasts may therefore assist with resource planning, public-health communication and intervention strategies.

This project is a comparative time-series study that evaluates two modelling approaches for describing and forecasting monthly influenza notifications in Australia. The first approach uses seasonal autoregressive integrated moving average (SARIMA) models, while the second uses exponential smoothing state-space (ETS) models. The project aims to determine which approach provides a more suitable model of historical influenza notifications and produces more accurate out-of-sample forecasts.

The project addresses the following research questions:

- Which model, SARIMA or exponential smoothing, produces more accurate out-of-sample forecasts of monthly influenza notifications?
- Do the selected SARIMA and ETS models produce residuals that are approximately uncorrelated, indicating that the main temporal structure in the data has been captured?

2 Data

The data used in this project were obtained from the Australian National Notifiable Disease Surveillance System (NNDSS) and cover the period from 1 January 2008 to 31 December 2019. The original dataset contains weekly influenza notifications, with weeks ending on a Friday. Cases are recorded according to state or territory, age group, sex, Indigenous status and influenza subtype.

Table 1: Summary of the final dataset used in the project.

Item	Detail
Data source	Australian National Notifiable Disease Surveillance System (NNDSS) influenza notifications
Sample period	1 January 2008 to 31 December 2019
Original frequency	Weekly, with weeks ending on a Friday
Final frequency	Monthly
Number of observations	144 monthly observations
Spatial coverage	Australian Capital Territory, New South Wales, Northern Territory, Queensland, South Australia, Tasmania, Victoria and Western Australia
Influenza types included	A(H3N2), A(H1N1)pdm09, A(H1N1), A(unsubtyped), B, C, A and B, and untyped cases
Aggregation	Total monthly influenza notifications across all states and territories, age groups, sexes, Indigenous-status categories and influenza subtypes
Outcome variable	Total number of monthly laboratory-confirmed influenza notifications
Missing data	None identified in the final aggregated dataset

For this project, the weekly records were aggregated to produce the total number of laboratory-confirmed influenza notifications per month. The data were not separated by state or territory, age group, sex, Indigenous status or influenza subtype. This aggregation was chosen because the aim of the project was to model and forecast national monthly influenza patterns.

It is important to note that notification data may underestimate the true number of influenza infections in the community. A case is included only if an individual seeks healthcare, is tested, receives a diagnosis and the case is subsequently reported to health authorities. The extent of underreporting is unknown and may vary over time, by jurisdiction and by influenza subtype.

After aggregation, the final dataset contained 144 monthly observations, covering January 2008 to December 2019. No missing values were identified in the final aggregated dataset.

3 Exploratory Data Analysis

The weekly influenza notification data were aggregated into monthly and yearly totals to examine broad patterns in influenza notifications over time. Figure **??** presents the monthly and yearly numbers of laboratory-confirmed influenza notifications in Australia from 2008 to 2019.

The monthly series shows a clear seasonal pattern. Influenza notifications generally increase during the middle of the year, reach a peak during the Australian winter months, and then decline towards the end of the year and the beginning of the following year. The yearly totals suggest an overall increase in reported influenza notifications over the study period, although this trend is not smooth. Large peaks are visible in 2009, 2017 and 2019, while the total number of notifications declined sharply in 2018.

The peak in 2009 coincides with the H1N1 influenza pandemic and may have been partly explained by the rapid spread of this strain. The high notification count in 2017 may have been influenced by factors such as circulating influenza strains, vaccine effectiveness, healthcare-seeking behaviour, testing practices and reporting patterns. Similarly, the increase in 2019 may have reflected differences in circulating strains, population immunity, vaccination coverage or surveillance practices. However, the NNDSS data used in this project do not include the additional information required to determine the specific causes of these increases. These explanations should therefore be regarded as possible contributors rather than confirmed causes.

Seasonal plots were used to assess whether the timing and magnitude of seasonality remained consistent over the study period. Figure **??** confirms that influenza notifications tend to rise during the middle of each year. However, the size of the seasonal peak differs across years. Earlier years, particularly from approximately 2010 to 2013, show relatively similar seasonal patterns, whereas the peaks in 2017 and 2019 are substantially larger. This suggests that the timing of influenza seasonality is fairly consistent, but the magnitude of the seasonal effect varies over time.

The subseries plot in Figure **??** provides a clearer view of the average monthly pattern. Influenza notifications begin to increase from approximately June, are highest on average in August, and then generally decline until October. This pattern is consistent with the timing of the Australian winter, which occurs mainly from June to August.

To separate the observed series into its underlying components, an STL decomposition was applied. Figure **??** shows the trend, seasonal and remainder components of monthly influenza notifications. The trend component is non-linear, with marked departures from the general pattern around 2009, 2017 and 2019. The seasonal component shows increasing variation over time, suggesting that the seasonal amplitude may be proportional to the level of the series. This indicates that the data may display multiplicative seasonal behaviour.

The remainder component fluctuates around zero for much of the series but contains large deviations in years with unusually high notification counts. These deviations indicate that events or factors other than the usual trend and seasonal pattern may have affected influenza notifications during those years.

Overall, the exploratory analysis provides evidence of strong annual seasonality, changing seasonal amplitude and a non-linear long-term pattern. These characteristics motivated the use of both SARIMA and ETS models.

4 Train–Test Split and Variance Stabilisation

To compare forecasting performance, the monthly influenza data were divided into training and test sets using an approximately 80:20 split. The training set covers January 2008 to December 2017 and contains 120 monthly observations. The test set covers January 2018 to December 2019 and contains 24 monthly observations.

This split was selected because the test set contains two complete annual seasonal cycles. It

therefore allows model performance to be evaluated across more than one influenza season. Keeping the final two years as the test set also provides an out-of-sample evaluation period that includes the substantial increase in notifications observed in 2019.

Before fitting SARIMA models, a variance-stabilising transformation was considered. The exploratory analysis suggested that the seasonal amplitude increased as the level of the series increased. This indicates multiplicative seasonality and suggests that the variance may not be constant over time.

The Box–Cox transformation is defined as

$$y_t^{(\lambda)} = \begin{cases} \dfrac{y_t^{\lambda} - 1}{\lambda}, & \lambda \neq 0, \\ \log(y_t), & \lambda = 0, \end{cases}$$

where y_t is the original number of influenza notifications in month t, $y_t^{(\lambda)}$ is the transformed value, and λ is the transformation parameter. When $\lambda = 1$, the data remain approximately on their original scale. When $\lambda = 0$, the transformation becomes the natural logarithm, provided that $y_t > 0$.

The optimal value of λ was estimated using the Guerrero method on the training data only. The test data were excluded from this step to prevent information leakage. Estimating the transformation parameter using only the training data ensures that the modelling procedure reflects the information that would have been available at the end of 2017.

The estimated Guerrero value was $\lambda = -0.07$, which is close to zero. A logarithmic transformation was therefore used as a simple and interpretable approximation to the Box–Cox transformation:

$$y_t^{(0)} = \log(y_t).$$

Figure **??** shows the transformed training series and its seasonal pattern. Compared with the original series, the seasonal peaks are more similar in size after transformation, suggesting that the variance has been stabilised. The transformed series still displays an upward movement over time, while the 2009 peak remains noticeably unusual.

An STL decomposition was then applied to the transformed training data. The trend component appears more gradual than in the original series, and the seasonal component shows a more consistent pattern across years. Although the remainder fluctuates around zero for much of the series, the large deviation around 2009 remains visible.

The transformed series was used for SARIMA modelling because the model requires the series to be made approximately stationary through transformation and differencing. The original series was retained for ETS modelling because ETS models can directly accommodate evolving level and seasonal components.

5 SARIMA Model Fitting

Seasonal autoregressive integrated moving-average (SARIMA) models extend ordinary ARIMA models by allowing both non-seasonal and seasonal patterns to be modelled. The autoregressive (AR) component models the current value of a time series using previous values. The moving-average (MA) component models the current value using previous forecast errors. The integrated (I) component refers to differencing, which can be used to remove non-stationary behaviour.

A SARIMA model is written as

$$\text{SARIMA}(p, d, q)(P, D, Q)_s.$$

The non-seasonal component, (p, d, q), captures short-term patterns in the series, while the seasonal component, $(P, D, Q)_s$, captures repeating patterns after each seasonal cycle. Since the data are monthly and influenza seasonality occurs annually, the seasonal period is

$$s = 12.$$

The general form of a SARIMA model is

$$\Phi_P(B^s)\phi_p(B)(1 - B)^d(1 - B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\varepsilon_t.$$

Here, y_t denotes the transformed influenza series at time t, B is the backshift operator and ε_t is a random error term. If the model is adequate, the error term should have a mean close to zero and no remaining autocorrelation.

The backshift operator is defined as

$$By_t = y_{t-1}.$$

For monthly data, the seasonal backshift is

$$B^{12}y_t = y_{t-12}.$$

The SARIMA parameters are defined as follows:

- p is the number of non-seasonal autoregressive terms.
- d is the number of non-seasonal differences.
- q is the number of non-seasonal moving-average terms.
- P is the number of seasonal autoregressive terms.
- D is the number of seasonal differences.
- Q is the number of seasonal moving-average terms.

- s is the number of observations in one seasonal cycle.

5.1 Stationarity Testing

SARIMA modelling requires the series to be approximately stationary after any necessary transformations and differencing. A stationary series has a stable mean and variance over time, and its dependence on previous observations does not change systematically through time.

The transformed training data were assessed using unit-root and stationarity tests. The unit-root tests were used to identify the number of non-seasonal and seasonal differences required. The KPSS test was also used to assess evidence against the null hypothesis that the series is stationary.

Table 2 shows the initial stationarity-test results. The results indicated that one seasonal difference and one non-seasonal difference may have been required. In addition, the small KPSS p-value provided evidence against the null hypothesis of stationarity.

Table 2: Initial stationarity-test results for the transformed training series.

Test or diagnostic	**Result**
Suggested non-seasonal differences, d	1
Suggested seasonal differences, D	1
KPSS test p-value	[Insert value]
Initial conclusion	Further differencing may be required

Seasonal differencing was applied first because the exploratory analysis showed strong annual seasonality. Seasonal differencing is defined as

$$(1 - B^{12})y_t = y_t - y_{t-12}.$$

The stationarity tests were then repeated on the seasonally differenced series. The results in Table 3 indicated that no further seasonal or non-seasonal differencing was required. The larger KPSS p-value indicated that there was insufficient evidence to reject the null hypothesis of stationarity.

Table 3: Stationarity-test results after applying one seasonal difference.

Test or diagnostic	**Result**
Suggested non-seasonal differences, d	0
Suggested seasonal differences, D	0
KPSS test p-value	[Insert value]
Conclusion	The seasonally differenced series was treated as stationary

Therefore, the candidate SARIMA models were fitted with one seasonal difference and no additional non-seasonal difference:

$$d = 0, \qquad D = 1.$$

5.2 Candidate Model Selection

Autocorrelation function (ACF) and partial autocorrelation function (PACF) plots of the seasonally differenced training series were used to identify plausible autoregressive and moving-average orders. The PACF showed significant non-seasonal spikes at lags one and two, suggesting a non-seasonal autoregressive order of two. The ACF and PACF also showed structure around the seasonal lag of twelve months, providing evidence for seasonal autoregressive or moving-average terms.

The initial baseline model was therefore specified as

$$\text{SARIMA}(2,0,0)(1,1,0)_{12}.$$

Because identifying AR and MA orders from ACF and PACF plots involves visual judgement, additional plausible models were fitted. An automatically selected ARIMA model was also fitted as an additional comparison.

Table 4 lists the candidate models used in the analysis.

Table 4: Candidate SARIMA models fitted to the transformed training data.

Model label	**SARIMA specification**	**Reason for inclusion**
Baseline	$\text{SARIMA}(2,0,0)(1,1,0)_{12}$	Suggested by the initial ACF and PACF patterns
Candidate 1	$\text{SARIMA}(2,0,1)(1,1,0)_{12}$	Adds a non-seasonal MA(1) term
Candidate 2	$\text{SARIMA}(2,0,1)(0,1,1)_{12}$	Tests a seasonal MA(1) term instead of seasonal AR(1)
Candidate 3	$\text{SARIMA}(2,0,0)(2,1,0)_{12}$	Tests an additional seasonal AR term
Automatic model	Selected using automatic ARIMA search	Provides a data-driven comparison

The models were compared using the Akaike Information Criterion (AIC), the corrected Akaike Information Criterion (AICc) and the Bayesian Information Criterion (BIC). Smaller values indicate a better balance between in-sample fit and model complexity.

Table 5: In-sample comparison of fitted SARIMA models.

Model	**Specification**	**AIC**	**AICc**	**BIC**
Baseline	$\text{SARIMA}(2,0,0)(1,1,0)_{12}$	[Insert]	[Insert]	[Insert]
Candidate 1	$\text{SARIMA}(2,0,1)(1,1,0)_{12}$	[Insert]	[Insert]	[Insert]
Candidate 2	$\text{SARIMA}(2,0,1)(0,1,1)_{12}$	[Insert]	[Insert]	[Insert]
Candidate 3	$\text{SARIMA}(2,0,0)(2,1,0)_{12}$	[Insert]	[Insert]	[Insert]
Automatic model	[Insert specification]	[Insert]	[Insert]	[Insert]

The model with the strongest in-sample fit was

$$\text{SARIMA}(2,0,0)(2,1,0)_{12}.$$

This model had the lowest AIC, AICc and BIC among the fitted SARIMA candidates. It was also consistent with the automatically selected ARIMA model, if the automatic procedure selected the same specification.

The selected model contains two non-seasonal autoregressive terms, no non-seasonal differencing, no non-seasonal moving-average terms, two seasonal autoregressive terms, one seasonal difference and no seasonal moving-average terms. Its model equation is

$$(1-\phi_1 B-\phi_2 B^2)(1-\Phi_1 B^{12}-\Phi_2 B^{24})(1-B^{12})y_t=\varepsilon_t.$$

The selected SARIMA model was subsequently assessed using residual diagnostics and out-of-sample forecasting performance.

6 Exponential Smoothing Models

The second family of models fitted to the influenza data was exponential smoothing state-space models. These models were considered because the exploratory analysis showed strong annual seasonality and a changing level over time. ETS models describe the series through evolving level, trend and seasonal components.

The notation for an ETS model is

$$\text{ETS}(E,T,S),$$

where E denotes the error component, T denotes the trend component and S denotes the seasonal component. Each component may be additive (A), multiplicative (M) or absent (N). The trend component may also be damped.

The exploratory analysis indicated that the size of the seasonal influenza peaks increased when the overall number of notifications was higher. Therefore, models with multiplicative seasonality were fitted. Models with additive and multiplicative errors were considered because the residual plots did not provide conclusive evidence about the error structure.

Simple exponential smoothing models were not explored because they contain neither a trend nor a seasonal component. Instead, models with additive, damped additive and absent trend components were considered.

The selected model was ETS(M,N,M), which has multiplicative errors, no explicit trend component and multiplicative seasonality. Its observation equation is

$$y_t=\ell_{t-1}s_{t-m}(1+\varepsilon_t),$$

where y_t is the observed value, ℓ_{t-1} is the level at time $t-1$, s_{t-m} is the seasonal index from

the previous seasonal cycle, $m = 12$ is the seasonal period and ε_t is the multiplicative error.

The level equation is

$$\ell_t = \ell_{t-1}(1 + \alpha\varepsilon_t),$$

and the seasonal equation is

$$s_t = s_{t-m}(1 + \gamma\varepsilon_t).$$

The h-step-ahead forecast is

$$\hat{y}_{t+h|t} = \ell_t s_{t+h-m(k+1)},$$

where k is the integer part of $(h-1)/m$. The parameters α and γ are the level and seasonal smoothing parameters, respectively.

The candidate ETS models are shown in Table 6.

Table 6: Candidate exponential smoothing models.

Model label	**ETS specification**	**Reason for inclusion**
Baseline	ETS(A, A, M)	Additive error and additive trend with multiplicative seasonality
Candidate 1	ETS(A, Ad, M)	Tests whether a damped additive trend improves the model
Candidate 2	ETS(M, A, M)	Tests multiplicative errors with an additive trend
Candidate 3	ETS(M, Ad, M)	Tests multiplicative errors with a damped additive trend
Candidate 4	ETS(M, N, M)	Tests whether multiplicative errors and seasonality are sufficient without an explicit trend
Candidate 5	ETS(A, N, M)	Tests additive errors with no trend and multiplicative seasonality
Automatic model	Automatically selected ETS model	Provides a data-driven comparison across eligible ETS specifications

The candidate models were compared using AIC, AICc and BIC. Lower values indicate a better trade-off between in-sample fit and model complexity.

Table 7: In-sample comparison of candidate exponential smoothing models.

Model label	**Specification**	**AIC**	**AICc**	**BIC**
Baseline	ETS(A, A, M)	[Insert]	[Insert]	[Insert]
Candidate 1	ETS(A, Ad, M)	[Insert]	[Insert]	[Insert]
Candidate 2	ETS(M, A, M)	[Insert]	[Insert]	[Insert]
Candidate 3	ETS(M, Ad, M)	[Insert]	[Insert]	[Insert]
Candidate 4	ETS(M, N, M)	[Insert]	[Insert]	[Insert]
Candidate 5	ETS(A, N, M)	[Insert]	[Insert]	[Insert]
Automatic model	[Insert specification]	[Insert]	[Insert]	[Insert]

Table 7 shows that Candidate 4, ETS(M, N, M), had the lowest AIC, AICc and BIC among the fitted ETS models. The automatic ETS procedure selected the same specification, if applicable. Therefore, the model with the strongest in-sample fit was

$$\text{ETS}(M, N, M).$$

This model has multiplicative errors, no explicit trend component and multiplicative seasonality. Although the original data appeared to show an overall increase in notifications, the information criteria suggested that including an explicit trend did not improve in-sample model fit once multiplicative seasonal variation was included.

7 Residual Diagnostics

Residual diagnostics were conducted for the selected SARIMA and ETS models to assess whether they adequately captured the temporal structure in the training data. A well-specified forecasting model should produce residuals that are approximately uncorrelated, have a mean close to zero and display no clear remaining pattern over time.

Figure **??** presents the residual diagnostics for the selected SARIMA model, SARIMA$(2, 0, 0)(2, 1, 0)_{12}$. The residual ACF plot contains several significant spikes, suggesting that some autocorrelation remains after fitting the model. In addition, the residual histogram appears right-skewed rather than symmetric around zero. These results suggest that the SARIMA model does not fully capture all of the temporal structure in the influenza series.

Figure **??** presents the residual diagnostics for the selected ETS model, ETS(M, N, M). The residual ACF plot shows fewer significant spikes than the SARIMA model, although at least one significant spike remains. The residual histogram appears left-skewed. Therefore, the ETS model also appears to retain some dependence that was not captured by the fitted components.

The Ljung–Box test was used to formally assess residual autocorrelation. The null hypothesis is that the residuals are independently distributed and contain no remaining autocorrelation. The alternative hypothesis is that autocorrelation remains in the residuals.

Table 8: Ljung–Box test results for the selected SARIMA and ETS models.

Model	**Lag**	**Test statistic**	***p*-value**
SARIMA$(2,0,0)(2,1,0)_{12}$	24	[Insert value]	[Insert value]
ETS(M,N,M)	24	[Insert value]	[Insert value]

As shown in Table 8, both models produced small p-values. Therefore, the null hypothesis of no residual autocorrelation was rejected for both models. This indicates that neither model completely captured the temporal dependence in the training data.

The remaining autocorrelation may be related to unusual influenza seasons, such as the large peaks observed in 2009, 2017 and 2019, or to external factors not included in the models. Possible omitted influences include changes in influenza strains, vaccination effectiveness, testing practices, reporting behaviour and weather conditions. However, the available dataset did not include these variables, and further investigation was outside the scope of this project.

Despite these diagnostic limitations, both models were retained for out-of-sample forecast evaluation. Their forecast accuracy over the 2018–2019 test period was compared with benchmark forecasting methods.

8 Forecasting and Model Comparison

8.1 In-Sample Model Fit

The selected SARIMA and ETS models were first examined using their fitted values on the training data. Parameter estimates were obtained using maximum likelihood estimation. The maximum likelihood procedure selects parameter values that make the observed training data most probable under the fitted model.

8.1.1 SARIMA Model

Based on the AIC, AICc and BIC comparison, the selected SARIMA model was

$$\text{SARIMA}(2,0,0)(2,1,0)_{12}.$$

The model equation is

$$(1-\phi_1 B-\phi_2 B^2)(1-\Phi_1 B^{12}-\Phi_2 B^{24})(1-B^{12})y_t=\varepsilon_t.$$

Here, y_t is the log-transformed monthly influenza count at time t, B is the backshift operator and ε_t is the random error term. The terms ϕ_1 and ϕ_2 represent the non-seasonal autoregressive coefficients at lags one and two months. The terms Φ_1 and Φ_2 represent the seasonal autoregressive coefficients at lags 12 and 24 months. The term $(1-B^{12})$ applies one seasonal difference to remove annual seasonal non-stationarity.

The maximum-likelihood parameter estimates are shown in Table 9.

Table 9: Parameter estimates for the selected SARIMA model.

Parameter	**Estimate**	**Interpretation**
ϕ_1	[Insert]	Non-seasonal AR coefficient at lag 1
ϕ_2	[Insert]	Non-seasonal AR coefficient at lag 2
Φ_1	[Insert]	Seasonal AR coefficient at lag 12
Φ_2	[Insert]	Seasonal AR coefficient at lag 24
Constant	[Insert, if included]	Constant or drift term, if included in the fitted model
Innovation variance, σ^2	0.08	Variance of the model errors
Log-likelihood	[Insert]	In-sample likelihood measure

The coefficient estimates indicate that the transformed influenza series is related to both recent observations and observations from previous influenza seasons. However, the signs and sizes of the coefficients should be interpreted with care. In a SARIMA model with seasonal differencing and multiple autoregressive terms, an individual coefficient does not establish a causal effect. For example, a negative seasonal autoregressive coefficient does not demonstrate that vaccination caused lower influenza counts. Instead, it indicates an inverse conditional relationship after accounting for the other model terms.

Figure **??** overlays the fitted SARIMA values on the transformed training data. The model captures the broad annual seasonal pattern. However, it appears to underestimate some periods between 2010 and 2011 and to show greater month-to-month variation than the observed series during non-seasonal periods.

8.1.2 ETS Model

The selected exponential smoothing model was

$$\text{ETS}(M, N, M).$$

This notation indicates multiplicative errors, no explicit trend component and multiplicative seasonality. The one-step-ahead observation equation is

$$y_t = \ell_{t-1} s_{t-12}(1 + \varepsilon_t),$$

where y_t is the observed monthly influenza count, ℓ_{t-1} is the level estimated at the previous time point, s_{t-12} is the seasonal index from the same month in the preceding year and ε_t is the multiplicative error term.

The level is updated as

$$\ell_t = \ell_{t-1}(1 + \alpha\varepsilon_t),$$

and the seasonal component is updated as

$$s_t = s_{t-12}(1 + \gamma\varepsilon_t).$$

The h-step-ahead forecast is

$$\hat{y}_{t+h|t} = \ell_t s_{t+h-12(k+1)},$$

where k is the integer part of $(h-1)/12$.
The parameter estimates are shown in Table 10.

Table 10: Parameter estimates for the selected ETS(M,N,M) model.

Parameter	**Estimate**	**Interpretation**
α	[Insert]	Level smoothing parameter
γ	[Insert]	Seasonal smoothing parameter
Innovation variance, σ^2	0.4631	Variance of the model errors
Log-likelihood	[Insert]	In-sample likelihood measure

The estimate of α was close to one, indicating that the estimated level responds strongly to recent observations. The estimate of γ was close to zero, suggesting that the seasonal pattern changes slowly over time and is influenced more by historical seasonal information than by the most recent observation. The model contains 12 initial seasonal states because the data have an annual seasonal cycle of 12 months.

Figure **??** overlays the fitted ETS values on the original training data. The model generally captures lower influenza-count periods, but it may overestimate or underestimate some major seasonal peaks. This is consistent with the residual diagnostics, which indicated that the model did not completely remove all autocorrelation.

8.2 Out-of-Sample Forecasting

The selected SARIMA and ETS models were used to forecast the test period from January 2018 to December 2019. Four benchmark forecasting methods were also included: mean, naive, seasonal naive and drift forecasts. These benchmark methods provide simple reference points against which the more complex models can be assessed.

Because the SARIMA model was fitted to the log-transformed series, its forecasts were back-transformed to the original influenza-count scale before comparison with the observed test data and the ETS forecasts. All reported accuracy measures were calculated using the original monthly counts.

Table 11 summarises the benchmark approaches.

Table 11: Benchmark forecasting methods.

Method	**Forecast rule**
Mean	Forecasts each future value using the mean of all training observations.
Naive	Uses the most recent training observation as the forecast for all future periods.
Seasonal naive	Uses the observation from the same month in the previous year.
Drift	Extends a straight line between the first and final training observations.

Figure **??** shows the benchmark forecasts against the observed test data. The seasonal naive forecast was visually closer to the observed data than the other benchmark methods, which is expected because the influenza data exhibit strong yearly seasonality. However, visual comparison alone is insufficient, so formal forecast accuracy measures were also used.

Figure **??** shows the SARIMA forecasts and their 80% and 95% prediction intervals. The intervals covered many observed test values, although the model appears to overestimate influenza counts during parts of 2018. The prediction intervals widen during high-season periods, reflecting greater uncertainty when forecasting influenza peaks.

Figure **??** shows the ETS forecasts and prediction intervals. The ETS model produced relatively narrow intervals during lower-incidence periods and wider intervals during seasonal peaks. This pattern reflects the greater uncertainty associated with forecasting periods in which influenza notifications are expected to increase substantially.

8.3 Forecast Accuracy

Forecast accuracy was evaluated using the root mean squared error (RMSE), mean absolute error (MAE) and mean absolute percentage error (MAPE). These measures are defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}.$$

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}\left|y_t - \hat{y}_t\right|.$$

$$\text{MAPE} = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right|.$$

Here, y_t is the observed influenza count, $\hat{y}_t$ is the forecast influenza count and n is the number of observations in the test set. Lower values indicate more accurate forecasts.

Table 12: Out-of-sample forecast accuracy for the 2018–2019 test period.

Model	**RMSE**	**MAE**	**MAPE**
Mean	[Insert]	[Insert]	[Insert]
Naive	[Insert]	[Insert]	[Insert]
Seasonal naive	[Insert]	[Insert]	[Insert]
Drift	[Insert]	[Insert]	[Insert]
SARIMA$(2,0,0)(2,1,0)_{12}$	[Insert]	[Insert]	[Insert]
ETS(M,N,M)	[Insert]	[Insert]	[Insert]

Table 12 shows that ETS(M,N,M) achieved the lowest RMSE and MAE, making it the most accurate model according to these two measures. The drift model obtained the lowest MAPE. This difference is possible because RMSE, MAE and MAPE weight forecasting errors differently. RMSE gives greater weight to large errors, MAE treats all absolute errors equally and MAPE measures error relative to the observed value.

Although the selected SARIMA model achieved the strongest in-sample fit among the SARIMA candidates, it performed worst during the out-of-sample forecast comparison. This demonstrates that low AIC, AICc or BIC values do not guarantee the best future forecasting performance. A possible explanation is that the SARIMA model's fitted dependence structure did not generalise well to the 2018–2019 period, particularly given the unusual behaviour of influenza notifications during that period.

Based on the out-of-sample results, the ETS(M,N,M) model was the best-performing model according to RMSE and MAE. However, the drift model performed best according to MAPE. Therefore, the conclusion that ETS was the best model should be stated with respect to the selected accuracy measures rather than as an unqualified claim.

9 Limitations and Reflection

Although the selected ETS(M,N,M) model achieved the lowest RMSE and MAE in the out-of-sample forecast comparison, it has important limitations. In particular, the selected model contains no explicit trend component. The exploratory analysis suggested that influenza notifications showed non-linear long-term variation, including substantial changes in particular years. Therefore, the selected model may not fully capture longer-term changes in influenza notification levels.

The unusually high influenza notification counts in 2009, 2017 and 2019 may also have affected model estimation and forecasting performance. These years contain observations that differ substantially from the usual seasonal pattern. Possible contributors include changes in circulating influenza strains, vaccination effectiveness, testing practices, healthcare-seeking behaviour and reporting. However, the dataset used in this project did not include variables that would allow these explanations to be tested directly.

The SARIMA and ETS residual diagnostics also indicated remaining autocorrelation. This suggests that time and seasonality alone were not sufficient to explain all variation in influenza

notifications. With additional time, future work could investigate models that include external predictors, such as temperature, rainfall, vaccination coverage, influenza subtype, testing rates or state-level information. Modelling influenza notifications separately by state or influenza subtype may also reveal patterns hidden by the national aggregation used in this project.

The analysis was deliberately restricted to data from 2008 to 2019. Data from 2020 onwards were excluded because the COVID–19 pandemic, public-health restrictions and changes in healthcare-seeking and testing behaviour could represent a structural break in the influenza series. Including this period without explicitly modelling such changes could distort the estimated seasonal patterns and lead to misleading forecasts.

Finally, the test set contained only 24 monthly observations. Although this provided two complete seasonal cycles, it is still a relatively short period for evaluating forecast performance. A longer dataset or rolling-origin forecast evaluation would provide stronger evidence about whether the ETS model consistently outperforms SARIMA and the benchmark methods.